

3.3 ggplot2: Facets, Themes, and Saving

Facet to add a category, set a theme and a scale, then `ggsave()` the object.

1. Facets

Facets make a panel for each level of a categorical variable. That is often clearer than twenty colors on one panel.

`facet_wrap(~ class)` wraps one variable into a grid:

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy)) +
  geom_point() +
  facet_wrap(~ class)
```

`facet_grid(rows ~ cols)` is a matrix of two categoricals:

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy)) +
  geom_point(alpha = 0.3) +
  facet_grid(drv ~ class)
```

A `.` means "no faceting on that side":

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy)) +
  geom_point() +
  facet_grid(class ~ .)

ggplot(data = mpg, mapping = aes(x = displ, y = hwy)) +
  geom_point() +
  facet_grid(. ~ drv)
```

2. A fixed color is not a mapping

To make every point blue, set `color` **outside** `aes()`:

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy)) +
  geom_point(color = "blue")
```

The name is a character, so it is in quotes. Inside `aes()`, `ggplot2` treats `"blue"` as a one-level categorical variable and draws a legend for it. That is a common mistake:

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy, color = "blue")) +
  geom_point()
```

The same rule applies to histograms. `fill` is the interior; `color` is the outline:

```
ggplot(data = mpg, mapping = aes(x = hwy)) +  
  geom_histogram(bins = 25, fill = "white", color = "black")
```

3. Themes

A theme sets the background, grid, and fonts in one step. `theme_bw()` is a usual choice for a document. `theme_classic()` is quieter.

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy, color = class)) +  
  geom_point() +  
  theme_bw()
```

`ggthemes` adds more, including `theme_economist()` and `theme_fivethirtyeight()`. Load the package first.

4. Scales and colorblind palettes

A scale maps data values to visual values. You can set the ends of a continuous color scale, or log both axes:

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy, color = cty)) +  
  geom_point() +  
  scale_color_continuous(low = "black", high = "white")  
  
ggplot(data = diamonds, mapping = aes(x = carat, y = price)) +  
  geom_point(alpha = 0.01) +  
  scale_x_log10() +  
  scale_y_log10()
```

For a plot you will share, use a colorblind-safe palette. `ggthemes::scale_color_colorblind()` and `scale_fill_colorblind()` are one option. `viridis` is another: colorful, perceptually even, and still readable in grayscale. Use `scale_color_viridis_d()` / `scale_fill_viridis_d()` for discrete groups and `_c` for continuous.

```
library(ggthemes)

ggplot(data = mpg, mapping = aes(x = displ, y = hwy, color = class)) +
  geom_point() +
  theme_bw() +
  scale_color_colorblind()

ggplot(data = mpg, mapping = aes(x = hwy, fill = drv)) +
  geom_histogram(position = "dodge", bins = 15) +
  theme_bw() +
  scale_fill_viridis_d()
```

5. Labels

`xlab()`, `ylab()`, and `ggtitle()` set the axis titles and the main title. The legend title is an argument on the scale (`name = ...`).

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy, color = class)) +
  geom_point() +
  theme_bw() +
  scale_color_discrete(name = "Car Class") +
  xlab("Displacement") +
  ylab("Highway") +
  ggtitle("Displacement vs Highway")
```

6. Saving a plot

A ggplot is an object. Assign it, print it, then export.

```
p1 <- ggplot(data = mpg, mapping = aes(x = displ, y = hwy, color = class)) +
  geom_point() +
  theme_bw()

p1
```

In RStudio, **Export** in the Plots pane writes a file. In a script or Quarto document, `ggsave()` does the same. It overwrites without asking.

```
ggsave("hwy-vs-displ.pdf", plot = p1, width = 6, height = 4)
```

If you omit `plot`, `ggsave()` uses the last plot you printed. Put the file in a folder you already have, or create one first.

7. This week, in one place

By the end of Week 3 you should be able to:

- build a plot with `ggplot()`, `aes()`, and a `geom_*()`
- choose a geom from the variables: scatter (two quantitative), histogram or density (one quantitative), boxplot (one categorical and one quantitative)
- map a third variable with `color`, `fill`, or facets
- add a smooth, fix overplotting, change a theme and a scale
- label the plot and save it with `ggsave()`

Functions from this week include `ggplot()`, `aes()`, `geom_point()`, `geom_smooth()`, `geom_jitter()`, `geom_hex()`, `geom_histogram()`, `geom_density()`, `geom_boxplot()`, `facet_wrap()`, `facet_grid()`, `theme_bw()`, `scale_y_log10()`, `xlab()`, `ylab()`, `ggtitle()`, and `ggsave()`.

Chapter 3 of [R for Data Science](#) and the [ggplot2 cheat sheet](#) are the usual references.

8. Practice

1. Boxplots of diamond `price` vs `cut`, log-y, faceted by `color`.
2. The same plot: orange boxes, `theme_bw()`.
3. Save that plot with `ggsave()`.